19. Initiation codon is
    a) UUU b) UGA
    c) AUG d) UAG

20. A eukaryotic gene contains two kinds of base sequences which of these plays an important role in protein synthesis?
    a) Introns b) Exons
    c) Both a and b d) None of the above

21. Codon – anticodon interactions occur by
    a) Covalent bond
    b) Electrostatic interactions
    c) Hydrogen bonds
    d) Hydrophobic interaction

22. Which of the following RNA polymerases is responsible for the transcription of protein coding genes in eukaryotes?
    a) RNA Pol I b) RNA Poly II
    c) RNA Pol III d) RNA Pol IV

23. How are RNA molecules transported out of the nucleus
    a) Passive diffusion through the membrane
    b) Through membrane pores in an energy independent process
    c) Through membrane pores in an energy dependent process
    d) Through a channel in the membrane that leads to the endoplastic reticulation

24. During translation the codon in mRNA is actually "read" by
    a) The A site in the ribosomes
    b) The P site in the ribosomes
    c) The anticodon in at RNA
    d) The anticodon is an amino acid

25. A complex of ribosome attached to a single strand of RNA is known as
    a) Polysome b) Polymer
    c) Polypeptide d) Okazaki fragment

26. Which of the following is the start codon
    a) AUG b) UGA
    d) UAA d) UAG

27. What is true about tRNA?
    a) It binds with an amino acid at its 3' end
    b) It has 5 double stranded regions
    c) It has a codon at one end which recognizes the anticodon of mRNA
    d) It looks like clover leaf in the three D structure

28. Which one of the following hydrolysis internal phosphodiester bonds in a polynucleotide chain?
    a) Lipase b) Exonuclease
    c) Endonuclease d) Protease

29. DNA element with ability to change position is called
    a) Cistron b) Transposon
    c) Intron d) Recon

30. Spliceosomes are not found in cells of
    a) Plants b) Fungi
    c) Animals d) Bacteria

31. During DNA replication Okazaki fragments are used to elongate
    a) The leading strand towards replication fork
    b) The lagging strand towards replication fork
    c) The leading strand away from replication fork
    d) The lagging strand away from replication fork

32. When two different genes came from same parent they tend to remain together.
    i) What is the name of this phenomenon?
    ii) Draw the cross with suitable example.
    iii) Write the observed phenotypic ratio.

33. If you cross dominant genotype PV/PV male *Drosophila* with double recessive female and obtain F_1 hybrid. Now you cross F_1 male with double recessive female.

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